



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 11

COMMON HALF YEARLY EXAMINATION 2024 - 25

MATHEMATICS

Time Allowed : 3.00 Hours]

[Max. Marks : 90

PART - I

20x1=20

1. Answer all the questions by choosing the correct answer from the given 4 alternatives
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark

1. If $A = \{(x, y) : y = \sin x, x \in \mathbb{R}\}$ and $B = \{(x, y) : y = \cos x, x \in \mathbb{R}\}$ then $A \cap B$ contains

- (1) no element (2) infinitely many elements
(3) only one element (4) cannot be determined.

2. The solution set of the following inequality $|x - 1| \geq |x - 3|$ is

- (1) $[0, 2]$ (2) $[2, \infty)$ (3) $(0, 2)$ (4) $(-\infty, 2)$

3. $\cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \dots + \cos 179^\circ =$

- (1) 0 (2) 1 (3) -1 (4) 89

4. The number of five digit telephone numbers having at least one of their digits repeated is

- (1) 90000 (2) 10000 (3) 30240 (4) 69760.

5. The remainder when 38^{15} is divided by 13 is

- (1) 12 (2) 1 (3) 11 (4) 5

6. If the equation of the base opposite to the vertex $(2, 3)$ of an equilateral triangle is $x + y = 2$, then the length of a side is

- (1) $\sqrt{\frac{3}{2}}$ (2) 6 (3) $\sqrt{6}$ (4) $3\sqrt{2}$

7. If $\Delta = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$, then $\begin{vmatrix} ka & kb & kc \\ kx & ky & kz \\ kp & kq & kr \end{vmatrix}$ is

- (1) Δ (2) $k\Delta$ (3) $3k\Delta$ (4) $k^3\Delta$

8. Vectors $\vec{a}$ and $\vec{b}$ are inclined at an angle $\theta = 120^\circ$. If $|\vec{a}| = 1$, $|\vec{b}| = 2$, then $[(\vec{a} + 3\vec{b}) \times (3\vec{a} - \vec{b})]^2$ is equal to

- (1) 225 (2) 275 (3) 325 (4) 300

9. At $x = \frac{3}{2}$ the function $f(x) = \frac{|2x-3|}{2x-3}$ is

- (1) continuous (2) discontinuous (3) differentiable (4) non-zero

10. If $f(x) = \begin{cases} ax^2 - b, & -1 < x < 1 \\ \frac{1}{|x|}, & \text{elsewhere} \end{cases}$ is differentiable at $x = 1$, then

- (1) $a = \frac{1}{2}, b = \frac{-3}{2}$ (2) $a = \frac{-1}{2}, b = \frac{3}{2}$ (3) $a = -\frac{1}{2}, b = -\frac{3}{2}$ (4) $a = \frac{1}{2}, b = \frac{3}{2}$

11. If $\int \frac{1}{x^2} dx = k \left(\frac{1}{3x} \right) + c$, then the value of k is

- (1) $\log 3$ (2) $-\log 3$ (3) $-\frac{1}{\log 3}$ (4) $\frac{1}{\log 3}$



12. Let A and B be two events such that $P(\overline{A \cup B}) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P(\bar{A}) = \frac{1}{4}$. Then the events A and B are
- (1) Equally likely but not independent
(2) Independent but not equally likely
(3) Independent and equally likely
(4) Mutually inclusive and dependent
13. $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x} =$
- (1) $\log ab$
(2) $\log \left(\frac{a}{b}\right)$
(3) $\log \left(\frac{b}{a}\right)$
(4) $\frac{a}{b}$
14. If $f(x) = x \tan^{-1} x$, then $f'(1)$ is
- (1) $1 + \frac{\pi}{4}$
(2) $\frac{1}{2} + \frac{\pi}{4}$
(3) $\frac{1}{2} - \frac{\pi}{4}$
(4) 2
15. If A and B are two events such that $A \subset B$ and $P(B) \neq 0$, then which of the following is correct?
- (1) $P(A/B) = \frac{P(A)}{P(B)}$
(2) $P(A/B) < P(A)$
(3) $P(A/B) \geq P(A)$
(4) $P(A/B) > P(B)$
16. The probabilities of occurrence of two events are respectively 0.21 and 0.49. The probability that both occur simultaneously is 0.16. Then the probability that none of the two occurs is
- (1) 0.14
(2) 0.46
(3) 0.30
(4) None of these
17. The total number of terms in the expansion of $(x+a)^{100} + (x-a)^{100}$ after simplification is
- (1) 50
(2) 202
(3) 51
(4) None of these
18. Angle between the lines represented by the equation $x^2 + 2xy \sec \theta + y^2 = 0$ is
- (1) 0
(2) 2θ
(3) $\frac{\theta}{2}$
(4) None of these
19. $\frac{\cos 10^\circ + \sin 10^\circ}{\cos 10^\circ - \sin 10^\circ} =$
- (1) $\tan 55^\circ$
(2) $\cot 55^\circ$
(3) $-\tan 35^\circ$
(4) $-\cot 35^\circ$
20. The value of $\lim_{x \rightarrow 0} \frac{e^x - e^{\tan x}}{x - \tan x}$ equals -
- (1) 0
(2) 1
(3) ∞
(4) $\frac{1}{2}$

PART - II

- Answer any 7 questions
- Each question carries 2 marks
- Question number 30 is compulsory

7 x 2 = 14

21. If $n(A) = 10$ and $n(A \cap B) = 3$, find $n((A \cap B)' \cap A)$.

22. Solve for x: $|3 - x| < 7$.

23. Evaluate: (i) 4P_4 (ii) 5P_3

24. Write the first 6 terms of the sequences whose nth terms are given below and classify them as arithmetic progression, geometric progression, arithmetico-geometric progression, harmonic progression and none of them: $\frac{3n-2}{3^{n-1}}$



25. Consider a hollow cylindrical vessel, with circumference 24 cm and height 10 cm. An ant is located on the outside of vessel 4 cm from the bottom. There is a drop of honey at the diametrically opposite inside of the vessel, 3 cm from the top. (i) What is the shortest distance the ant would need to crawl to get the honey drop? (ii) Equation of the path traced out by the ant. (iii) Where the ant enters into the cylinder?

26. Construct an $m \times n$ matrix $A = [a_{ij}]$, where a_{ij} is given by $a_{ij} = \frac{|3i-4j|}{4}$ with $m = 3, n = 4$

27. Find a unit vector along the direction of the vector $5\hat{i} - 3\hat{j} + 4\hat{k}$.

28. Using the table estimate the value of the limit: $\lim_{x \rightarrow 2} \frac{x-2}{x^2-x-2}$

x	1.9	1.99	1.999	2.001	2.01	2.1
f(x)	0.344820	0.33444	0.33344	0.333222	0.33222	0.332258

29. If A and B are mutually exclusive events $P(A) = \frac{3}{8}$ and $P(B) = \frac{1}{8}$, then find $P(A \cup B)$

30. Evaluate: $\int a^{3x+3} dx$

PART - III

1. Answer any 7 questions

2. Each question carries 3 marks

3. Question number 40 is compulsory

7 x 3 = 21

31. Let $X = \{a, b, c, d\}$ and $R = \{(a, a), (b, b), (a, c)\}$. Write down the minimum number of ordered pairs to be included to R to make it (i) reflexive (ii) symmetric (iii) transitive (iv) equivalence

32. Differentiate $(2x+1)^5(x^3-x+1)^4$

33. Prove that $\tan\left(\frac{\pi}{4} + \theta\right) \tan\left(\frac{3\pi}{4} + \theta\right) = -1$.

34. The AM of two numbers exceeds their GM by 10 and HM by 16. Find the numbers.

35. Rewrite $\sqrt{3}x + y + 4 = 0$ in to normal form.

36. Without expanding the determinants, show that $|B| = 2|A|$. Where $B = \begin{bmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{bmatrix}$ and $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$

37. For any vector $\vec{a}$ prove that $|\vec{a} \times \hat{i}|^2 + |\vec{a} \times \hat{j}|^2 + |\vec{a} \times \hat{k}|^2 = 2|\vec{a}|^2$.

38. Integrate with respect to x: $\frac{e^{2x}-1}{e^x}$

39. A problem in Mathematics is given to three students whose chances of solving it are $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{5}$

(i) What is the probability that the problem is solved? (ii) What is the probability that exactly one of them will solve it?

40. Find the co-efficient of x^4 in the expansion $\frac{3x}{(x-2)(x+1)}$.



1. Answer all the questions

2. Each question carries 5 marks

41. a) Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = 2x - |x|$ and $g(x) = 2x + |x|$. Find $f \circ g$.

(OR)

b) Resolve the rational expressions into partial fractions: $\frac{(x-1)^2}{x^3+x}$

42. a) Prove that $\frac{\cot(180^\circ + \theta) \sin(90^\circ - \theta) \cos(-\theta)}{\sin(270^\circ + \theta) \tan(-\theta) \operatorname{cosec}(360^\circ + \theta)} = \cos^2 \theta \cot \theta$

(OR)

b) State and prove Napier's Formula.

43. a) Prove that for any natural number n , $a^n - b^n$ is divisible by $a - b$, where $a > b$

(OR)

b) Find the constant term of $\left(2x^3 - \frac{1}{3x^2}\right)^5$.

44. a) A photocopy store charges ₹1.50 per copy for the first 10 copies and ₹1.00 per copy after the 10th copy. Let x be the number of copies, and let y be the total cost of photocopying. (i) Draw graph of the cost as x goes from 0 to 50 copies. (ii) Find the cost of making 40 copies

(OR)

b) Prove that the points whose position vectors $2\hat{i} + 4\hat{j} + 3\hat{k}$, $4\hat{i} + \hat{j} + 9\hat{k}$ and $10\hat{i} - \hat{j} + 6\hat{k}$ form a right angled triangle.

45. a) Show that $\begin{vmatrix} b+c & a & a^2 \\ c+a & b & b^2 \\ a+b & c & c^2 \end{vmatrix} = (a+b+c)(a-b)(b-c)(c-a)$

(OR)

b) Evaluate: $\lim_{x \rightarrow \infty} \left(\frac{x+2}{x-2}\right)^x$

46. a) If $y = \tan^{-1} \left(\frac{1+x}{1-x}\right)$, find y'

(OR)

b) If $u = \tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ and $v = \tan^{-1} x$, find $\frac{du}{dv}$.

47. a) Integrate the function with respect to x : $\frac{8^{1+x} + 4^{1-x}}{2^x}$

(OR)

b) There are two identical urns containing respectively 6 black and 4 red balls, 2 black and 2 red balls. An urn is chosen at random and a ball is drawn from it. (i) find the probability that the ball is black (ii) If the ball is black, what is the probability that it is from the first urn?